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COMMENTARY

Cultural differences in social communication and interaction: A gap in autism research

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Abstract

Social communication and interaction deficits are a diagnostic criteria of autism and integral to practitioner and researcher conceptualization. Culture is an influential factor in expectations for, and demonstration of, social communication and interaction skills, but there is limited research published in autism journals on this topic. This paucity of autism research examining cultural factors related to social communication and interaction may contribute to known identification disparities for racial and ethnic populations minoritized by systemic factors and research bias. We call for increased commitment from researchers to recruit racially and ethnically minoritized participants, prioritize investigating cultural expectations and perceptions of social communication and interaction, and evaluate measures related to social communication for cultural and linguistic responsivity.

Lay Summary

A diagnosis of autism requires the presence of deficits in social communication and interaction. Examples of these behaviors and skills include holding a back-and-forth conversation, the use of nonverbal communicative behaviors (e.g., gestures), and developing and maintaining social relationships. Culture influences the expectations for, and presentation of, these behaviors. However, research on this topic is lacking. Conducting more research related to culture and social communication could help reduce the disparities in autism identification across racially and ethnically minoritized populations.

KEYWORDS
culture, race and ethnicity, social communication

INTRODUCTION

Deficits in social communication and interaction (SC/I) skills are one of the central characteristics of autism (American Psychiatric Association [APA], 2013). Broadly defined, SC/I is the use and integration of nonverbal and verbal skills, including both receptive and expressive language, for the purpose of social interaction (Mitchell et al., 2006; Stone et al., 1997). SC/I skills are important due to their contribution to one's ability to successfully engage in daily interactions and understand social exchanges. Because deficits in SC/I can impede effective communication, children with autism may have difficulties making and maintaining friends, be subjected to bullying and peer exclusion (Bauminger, 2002; Ingersoll & Wainer, 2013; Sofronoff et al., 2011), have an increased risk for depression and social anxiety (Segrin & Flora, 2000), and experience future occupational difficulty (e.g., Ke et al., 2017). Although SC/I skills are an integral component to an autism diagnosis, limited research has been published regarding cultural differences in SC/I across race and ethnicity. This paper reviews cultural differences in SC/I expectations and developmental concerns, cultural limitations of autism measures with SC/I components, limited representation of racially and ethnically minoritized (REM) populations in SC/I research, and resulting diagnostic disparities. We conclude the commentary advocating for increased research in this domain and actionable steps to achieve that goal.

CULTURAL DIFFERENCES IN SOCIAL COMMUNICATION

The importance of SC/I skills, their impact on everyday interactions and general functioning, and related judgments of quality are influenced by culture, an element not always explicitly captured in autism evaluations. The subjective perception of skills and deficits related to SC/I is derived from various norms and expectations within a culture (de Leeuw et al., 2020). Culture is a complex and overarching phenomenon that influences group and individual behaviors, values, and ideals (Birukou et al., 2013). Variations within cultural groups are common and further complexified by various intersecting cultural identities as culture is highly heterogenous. Culture encompasses and is expressed through language, social norms and rites, beliefs, relationships, and other customs (Rathje, 2009) and is a defining factor in the ways people communicate and interact with one another. Since the presence of SC/I deficits are core criteria for autism identification, culture should be used as a frame of reference when assessing and interpreting related skills and deficits. This commentary focuses on issues related to race and ethnicity, though we recognize other aspects of culture deserve further study and interpretation.

Culture clearly influences nonverbal communication (Marsh et al., 2003), one of the three sub-criteria related to SC/I in autism (APA, 2013). For example, avoiding eye contact is a common symptom of autism, although reduced eye contact may also be a sign of respect in Native American and East Asian cultures (Akechi et al., 2013; Uono & Hietanen, 2015). Cultural differences can also be seen in preferences for interpersonal space during interaction; some individuals from East Asian cultures, like Japan, may prefer greater interpersonal distance in conversation than do individuals from Western cultures (Sicorello et al., 2019). These differences may impact autism evaluations in that Western evaluators may interpret this need for greater space during conversation as a lack of engagement in social communication. Frequency and timing of gestures also differ from culture to culture. While individuals from Western countries may often use more representational and nonrepresentational gestures than those from East Asian cultures, variation also exists within Western cultures as well (So, 2010). For example, Italians gesture more frequently overall and are more likely to gesture while listening during a conversation than British individuals (Cavicchio & Kita, 2013). Even pointing, an SC/I behavior considered in autism evaluations in early childhood, is subject to cultural variation. Some individuals from different cultures show preference for pointing with the lips (e.g., Nigeria, Brazil, Malaysia; Mihas, 2017), nose (e.g., Papua New Guinea; Cooperrider & Núñez, 2012), chin (e.g., China; Yang, 2010) or with the index finger (e.g., United States, Germany; Cooperrider et al., 2018). When evaluators and clients are from cultures that have different expectations for gestures, misinterpretation that influences diagnostic conceptualization may occur.

Expressing and interpreting emotions are common difficulties for children with autism and are relevant to both nonverbal communication and making and maintaining relationships. However, culture also affects emotional expression and interpretation (Yuki et al., 2007). Specific to emotional expression, cultures have differing display rules dictating what emotions may be shared and how. An international study of emotion display rules (Matsumoto et al., 2008) illustrates that differences in emotion expressiveness are correlated with a culture's individualism. Cultures high in individualism (e.g., United States, Denmark, Australia) are typically more emotionally expressive than cultures low in individualism (e.g., Indonesia, Hong Kong, Russia). Of course, Matsumoto et al. (2008) found exceptions to this, such as Zimbabwe—a country with low individualism and high emotional expressiveness. Interestingly, this dichotomy may not hold true when people are interacting with others from their own culture as compared to those from different cultures. When interacting with people from the same culture, individuals are more expressive than predicted by display rules than when interacting with out-group members (Matsumoto et al., 2008). This final finding poses significant implications for cross-cultural evaluations for autism where children from cultures that are different than the practitioner may be perceived as less expressive.

There are also differences between Eastern and Western cultures in the interpretation of emotion. Many people from Asian cultures may focus on the eyes to interpret emotions while many Americans might focus on the mouth (Yuki et al., 2007). It may be difficult for someone to interpret the emotions of an individual from another culture since people rely on the display rules and interpretation cues of their own culture (Koelkebeck et al., 2015). This may be especially challenging for children with autism that are more likely to have difficulty identifying emotions generally (APA, 2013). Interestingly, American children with autism demonstrate greater accuracy in identifying facial expressions when they focus on the eyes (Bal et al., 2010), contradictory to the aforementioned patterns seen in neurotypical samples. However, emotion identification research in children with autism is limited in its representation of other cultures.

Verbal social communication, another skill set that may be difficult for children with autism, is also subject to cultural variation. In addition to more obvious differences in linguistics, culture influences the social norms and behaviors people use to communicate. For example, collectivist cultures may use more indirect communication, focusing on listening to others (Mio et al., 2019), and utilize high-context conversation which relies on implicit or indirect meanings related to shared knowledge of culture and context (Gudykunst et al., 1996). Since common characteristics of autism include difficulty conducting a back-and-forth conversation or the appearance of not listening to others when they talk, children with autism in collectivist cultures may face additional social impairment. On the other hand, individualistic cultures are more often associated with direct communication where message delivery is prioritized (Mio et al., 2019). People from individualistic cultures are more likely to use low-context communication styles, which are more direct and explicit in their sharing of ideas (Gudykunst et al., 1996). This method of communication may be easier to navigate than other cultural communication styles for people with autism. Due to the clear impact of culture on SC/I, culture should be considered an integral piece of the autism evaluation process (Matson et al., 2017).

Awareness of cultural differences in SC/I is of increasing importance in both traditionally racially homogenous countries as well as countries with greater racial and ethnic diversity. United States (U.S.) Census data indicates racial and ethnic diversity is increasing in the country. The proportion of U.S. residents identifying as White has decreased by 8.6% since 2010, while the proportion of those identifying as multiracial or Black increased by 276% and 129%, respectively (U.S. Census, 2020). Importantly, children and adolescents, who are most likely to receive an autism evaluation, demonstrate more racial and ethnic diversity than adult populations (U.S. Census, 2020). Impacts of globalization and increased immigration suggest that other countries, particularly those in Western Europe, are currently or will in the future experience increases in racial and ethnic diversity (Dao et al., 2018). As racial and ethnic diversity increases, if the production and consumption of culturally-responsive research does not keep pace, practitioners are at risk of failing to identify children and adolescents with autism whose cultures may differ from their own.

A growing awareness and acknowledgement of systemic racism (Broder-Fingert et al., 2020) and implicit bias (Obeid et al., 2020) in the field of autism further highlight the necessity of this research. Practitioners unaware of cultural differences in SC/I (e.g., social initiation and response patterns, play, use of gestures) may inappropriately identify a child as having autism. Lack of racial match between practitioners providing diagnosis and children receiving diagnosis is rampant. In the U.S., approximately 65% of pediatricians (Goodman, 2005) and 86% of psychologists (U.S. Census Bureau, 2015) identify as White. Limited recruitment and retention of REM professionals to diagnose autism coupled with a lack of research to aid White-identifying providers in culturally-responsive care limits the possibility of appropriate identification for REM children and adolescents.

RACE, ETHNICITY, CULTURE, AND AUTISM

Caregivers of children with autism from different racial and ethnic backgrounds show differential concern when their child displays potential autism symptoms, which may be based on cultural values. For instance, while caregiver concerns related to SC/I deficits are associated with earlier evaluation for autism (Becerra-Culqui et al., 2018), the domains of concern within SC/I vary. White caregivers may be more concerned about their child's language delays and verbal communication (Coonrod & Stone, 2004; Matson et al., 2017) whereas Latinx caregivers may be more concerned about their child's emotion regulation (Ratto et al., 2016). Samadi (2020) found Iranian parents reported significant concern about their child's understanding of social context and relationships with others; however, the majority did not feel concern until their child was 3 years of age. Culture may also impact how caregivers perceive autism symptom severity due to varying child development expectations (Matson et al., 2017). Interpretation of SC/I deficits also varies by culture. For example, in South Korea, SC/I difficulties may result in a diagnosis of reactive attachment disorder because it is more socially acceptable than autism as mothers are often blamed for autism (Kim, 2012).

Differing developmental and cultural expectations are problematic as the majority of diagnostic measures for autism are normed on fairly homogenous samples representing predominately White, Western individuals with minimal guidance on how practitioners can increase culturally responsive assessment practices (Harris et al., 2014). Stronach and Wetherby (2017) found that Black and Latinx caregivers reported more autism symptoms and SC/I deficits than White caregivers. Additionally, Black and Latinx children had significantly lower scores on speech-language observation measures than White children. Whether this is due to cultural differences in language expression or examiner scoring bias is unclear.

A review of autism diagnostic and screening measures found insufficient information was provided by test publishers regarding race and ethnicity and subsequent recommended considerations and practices, which can potentially influence administration and interpretation of these measures (Harris et al., 2014). For example, although the Autism Spectrum Rating Scales (ASRS) does not exhibit mean standard score differences across raters from different racial and ethnic groups (Goldstein & Naglieri, 2010), recent research suggests potential non-invariance at both the test (McClain et al., 2020b) and item levels (McClain et al., 2020a). Similar non-invariance occurs in the ADOS-2 across child race/ethnicity such that Black and Latinx children with the same behavioral presentations of autism consistently received higher practitioner ratings for poor eye contact, idiosyncratic speech, and echolalia (Harrison et al., 2017). Additionally, Albores-Gallo et al. (2012) found cultural differences in responding to critical items on the Modified Checklist for Autism in Toddlers (MCHAT) (Robins et al., 2001) by Mexican parents. These differences may result in inappropriate interpretation of measures or failure to appropriately identify autism. Furthermore, since the MCHAT is an autism screening instrument, differences in interpretation of this measure by both the caregiver and the professional may contribute to delays in obtaining recommended comprehensive autism assessment.

This skew in measure development research is additionally problematic in the global context. While most autism measures are developed and normed in the U.S. (Harris et al., 2014), when exported to other countries, limited modifications to adapt to the cultural norms and normative differences in the new location are made. As Ruparelia et al. (2016) have noted, many delegates at an international neurology conference have advocated for entirely new screening and diagnostic measures suited to the cultural and clinical realities of the African countries they represent. The use of a new measure, Hiva, developed in Iran was more efficacious in screening autism than Western measures (Samadi & McConkey, 2015). Fortunately, some existing measures have been translated and re-validated to other languages including Brazilian Portuguese (e.g., Becker et al., 2012) and Afrikaans (e.g., Smith et al., 2017). Unfortunately, however, these adaptations are often limited as they are only translated and back translated without consideration of cultural differences related to SC/I skills and parent expectations (Pereira et al., 2008). In South Korea, initial steps to translate and validate the ADOS-2 have shown promise, with reduced cut-off scores adjusted to the cultural norms and expectations (Lee et al., 2019). Until culturally valid measures are internationally available, many populations may lack the resources to both appropriately identify autism and produce autism research.

The lack of research about potential performance differences among cultural groups on autism assessments and limited culturally responsive measures for autism may influence the skew in diagnostic rates between REM populations and their White counterparts. In a review of representation in autism research only 21.5% of studies reported race/ethnicity demographic data—among those studies which included this information, White participants were overrepresented (Harris et al., 2020). The lack of racial and ethnic diversity and reporting in autism research is compounded by systemic issues faced by REM groups, particularly Black families, to access autism evaluations (Broder-Fingert et al., 2020; Constantino et al., 2020). Broader issues within academia may also impact this skew in research. For example, research with cultural applications is not always published in international journals or translated out of its original language (e.g., Manzone, 2013), making it difficult for providers in other countries to consume. Behavioral research, not just research on autism, also suffers from the myth that Western or White samples bear the benefit of universal generalizability while samples contextualized in other countries or cultures do not carry this misconception (Cheon et al., 2020). As such, racial and ethnic composition of samples may not be reported or may disproportionately represent White populations, limiting our understanding of autism and SC/I in other populations.

Sell et al. (2012) noted conflicting evidence in the literature regarding autism diagnostic disparities in regions of the U.S. In an evaluation of three cohorts of children living in the Philadelphia area, the researchers found no significant difference between the number of Black children and White children that met the criteria for an autism diagnosis, nor did race impact the presence of externalizing symptoms. However, a national monitor of autism prevalence indicates REM populations such as Latinx children are identified with autism less often than their White peers (Maenner et al., 2020). Further, although there is no difference in timing of caregiver concerns about autism symptoms across racial groups (Issarraras et al., 2019), REM children are diagnosed with autism, on average, several years later than their White peers (Mandell et al., 2009). This disparity in timing of diagnosis bars groups from access to early intervention services, which are linked to numerous positive long-term outcomes for children with autism (Camarata, 2014).

SUMMARY AND CONCLUSIONS

Cultural differences are often overlooked in understanding SC/I in the autism community. These differences are practically important for practitioners when evaluating and diagnosing children with autism. An accurate picture of cultural influences is crucial to decreasing diagnostic disparities, improving the cultural responsivity of autism measures, and increasing practitioner capacity to identify autism in REM populations. As such, we call for increased research related to race, ethnicity, and culture and their role in SC/I. The lack of research in this domain has been longstanding, and systemic changes are recommended to incentivize and enable investigations in this area. For example, grant funding agencies can exert their influence by amending their existing grants or developing new grants that fund projects that include international collaborations, take community-based participatory research approaches to include minoritized races and cultures, and intentionally include REM groups in samples.

Research dissemination outlets such as journals and conferences are ideally placed to emphasize and bolster these efforts. Increased acceptance and active solicitation of qualitative research, such as community-based participatory approaches, can uplift the voices and priorities of REM groups in autism and SC/I research. Within the review process, journals and conferences should also prioritize the critical examination of inclusion or exclusion of REM populations within submissions of all types, not just submissions for special diversity issues or presentations. Just as some journals (e.g., Autism) have begun requiring statements justifying the inclusion or exclusion of autistic community members in all publications, editorial boards may consider requiring these considerations to promote inclusion of REM groups. At the institutional level, renewed efforts to recruit, train, and retain researchers from REM populaions can help to promote and collaborate in these efforts to generate this research. Leadership of professional organizations related to autism should also seek to better reflect international and REM populations involved in their fields and modify or create policies to encourage relevant research and appropriate, culturally-responsive practice based on the findings.

Finally, researchers also maintain responsibility to generate this work. Future research should prioritize investigations of cultural factors related to autism and SC/I. Researchers should use current research regarding cultural differences in SC/I when designing studies and analyzing data by selecting measures that align with the target culture's norms for SC/I or adapting measures accordingly and thoroughly, and/or statistically controlling for cultural differences when warranted by the research question. Further understanding of SC/I contextualized within and across cultures can be used to improve assessment tools and strategies. Both researchers and assessment publishers should prioritize cultural considerations when developing or adapting measures through use of measurement invariance analysis, focus groups, and other robust methods that comprehensively consider race, ethnicity, and culture.

CONFLICT OF INTEREST

The authors declare no potential conflict of interest.

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